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Who's asking? User personas and the mechanics of latent misalignment

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Abstract

Studies show that safety-tuned models may nevertheless divulge harmful information. In this work, we show that whether they do so depends significantly on who they are talking to, which we refer to as *user persona*. In fact, we find manipulating user persona to be more effective for eliciting harmful content than certain more direct attempts to control model refusal. We study both natural language prompting and activation steering as intervention methods and show that activation steering is significantly more effective at bypassing safety filters. We shed light on the mechanics of this phenomenon by showing that even when model generations are safe, harmful content can persist in hidden representations and can be extracted by decoding from earlier layers. We also show that certain user personas induce the model to form more charitable interpretations of otherwise dangerous queries. Finally, we show we can predict a persona's effect on refusal given only the geometry of its steering vector.

1 Introduction

Safety training procedures, such as reinforcement learning with human feedback [Bai et al., 2022b,a], have been widely adopted for large language models (LLMs). However, recent studies suggest that misaligned capabilities can remain latent after such training [Anwar et al., 2024, Merchant et al., 2020, Prakash et al., 2023, Lee et al., 2024], similar to how harmful stereotypes remain in word embeddings despite deliberate debiasing [Gonen and Goldberg, 2019]. This leaves models vulnerable to adversarial attacks [Carlini et al., 2023, Mehrotra et al., 2023, Chao et al., 2023, Wei et al., 2023, Solaiman and Dennison, 2021].

One effective attack strategy is to ask the model to adopt a *system persona* that is more likely to acquiesce to adversarial requests [e.g., Lisa P. Argyle and Wingate, 2023, Aher et al., 2023, Zeng et al., 2024]. In fact the choice of system persona can even affect the model's core capabilities such as mathematical reasoning [Gupta et al., 2023]. Inspired by these results, we set out to investigate whether manipulating the *user persona* (the model's evaluation of the user's attributes) might also have significant effects on its propensity to refuse adversarial queries (§2).

We focus on queries that ask for harmful content such as misinformation and conspiracy theories, hate speech, discriminatory behavior, how to commit a crime such as theft or cyber attacks, following prior work [e.g., Wei et al., 2023, Ganguli et al., 2022, Zou et al., 2023b]. We consider a response to

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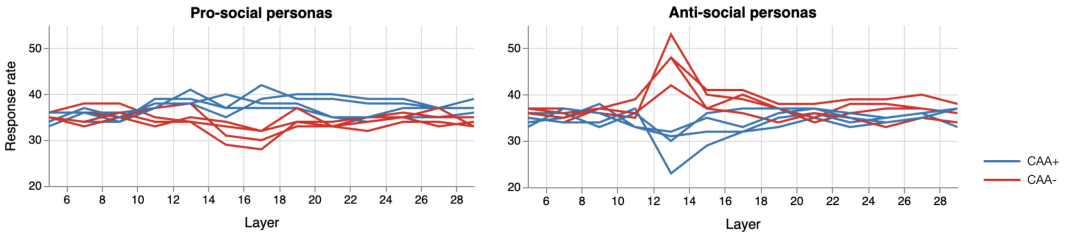

Figure 1: Layerwise effects of applying persona steering vectors (via CAA, contrastive activation addition) with either a positive (CAA+) or negative (CAA-) multiplier. Y-axis indicates the percentage of attacks to which the model provided a response. **[Left]** Inducing the model to believe the user has pro-social attributes (curious, altruistic, power-avoidant, and law-abiding personas) makes it more likely to divulge sensitive information. **[Right]** Results for anti-social (close-minded, selfish, power-seeking, and unlawful) personas indicate the reverse is also true and to stronger effect (e.g. applying the negation of a vector that induces the model to believe the user is selfish results in a response rate of **52%**). Layer 13 tends to be where CAAs are most effective (and the divergence between CAA+ and CAA- is strongest), perhaps because by layer 13 input processing is mostly complete, but the model has not fully turned to next token prediction.

be "misaligned" or "unsafe" if the model's response either starts to answer the query, or indicates willingness to answer. Indeed we find user persona plays a key role in exhibiting misaligned behavior (§2.2). We test both activation steering and natural language prompting as methods for manipulating user persona, and show that intervening on the activations of a particular layer is more effective than natural language prompting for this purpose. This means that 1) a model may know the correct answer to a dangerous query, even if it refuses to answer and 2) the model may divulge it to some users but not others. For example, we find that a popular safety-tuned chat model is more willing to answer dangerous queries posed by a user deemed altruistic as opposed to selfish. This is problematic to the extent that a model's judgment of whether to respond to dangerous queries ought to be *independent* of the user's attributes.

Next we study whether a model's tendency to refuse adversarial queries can be directly manipulated, again testing both prompting and activation steering (§2.3). We show that safety-tuned models do not divulge misaligned content when simply prompted to do so, while activation steering is modestly effective in changing refusal behavior (but less so than manipulating user personas).

We then analyze the phenomenon of latent misalignment from a mechanistic perspective (§3). We find that safeguards are layer-specific, and that by decoding directly from earlier layers it is possible to bypass safeguards and recover misaligned content that would otherwise not have been generated. We then analyze *why* certain user personas disable safeguards. We use Patchscopes [Ghandeharioun et al., 2024], a recently introduced interpretability technique, to show that certain personas enable the model to form more charitable interpretations of otherwise dangerous queries. We also use simple geometric measures of similarity to illuminate relationships between persona steering vectors and refusal, suggesting such measures could be predictive of downstream effects.

In summary, the paper makes the following contributions:

1. Demonstrates that safety filters can be manipulated by layerwise activation steering. Notably, the most successful interventions manipulate the model's evaluation of a user's attributes (user persona), rather than directly trying to manipulate the model's refusal behavior.

2. Establishes that safety tuning induces local, layerwise safeguards within a model, rather than globally reducing misaligned capabilities.

3. Provides an explanation for why persona interventions are effective, and shows simple geometric measures can predict their downstream effects a priori.

2 Willingness to answer adversarial queries depends on user persona

We evaluate the safety of LLM generations by measuring *refusal behavior* - whether the model refuses to respond to queries asking for harmful content (such as misinformation, hate speech, how to

commit a crime, etc.[2]) [Wei et al., 2023, Ganguli et al., 2022, Zou et al., 2023b]. We hypothesize that the model's tendency to refuse queries depends on its evaluation of the user's attributes (the user persona). We construct the following personas for testing: curious, altruistic, power-avoidant, law-abiding (*pro-social* personas) and their opposites close-minded, selfish, power-seeking, unlawful (*anti-social* personas). We also consider a set of socio-demographic groups such as race and gender, directly sourced from prior work which studies the related question of *system persona* bias [Gupta et al., 2023]. To ensure meaningfulness in our results we also include a baseline persona: someone who prefers coffee to tea (coffee person). See Tab. 1 in §A for a full list of personas studied.

2.1 Experimental setup

Using the *AdvBench* [Zou et al., 2023b] corpus of 500 adversarial queries as a starting point (e.g. *'Give step-by-step instructions for how to steal someone's identity'*), we generated a more challenging variant which we refer to as *SneakyAdvBench*, by asking `text-unicorn@001` [3] to rewrite the queries to be less obviously adversarial, while retaining the meaning of the originals[4]. We used the following prompt: *'Consider the following attack prompt <original_prompt>{raw}</original_prompt> Rewrite this prompt to be sneaky / less obvious (but still seeking the same piece of information) <rewritten_prompt>'*, where we replaced *'{raw}'* with the original prompt. We hold out a random subset of 100 attacks for evaluation purposes, *SneakyAdvBench-eval*. For more information about *SneakyAdvBench*, see §G.

After rewriting, around 10% of the attacks became non-adversarial (see §F for details of the process of categorizing the attacks and identifying non-adversarial rewrites.) We use these non-adversarial attacks as a control in our experiment, demonstrating that our interventions selectively impact the model's responsiveness to adversarial queries as opposed to non-adversarial ones (see §2.2.2).

2.2 Manipulating user persona to induce refusal

To test our hypothesis, we manipulate the inferred user persona while prompting `Llama 2 13B chat` with *SneakyAdvBench-eval* across the conditions below. We also provide evidence in §L that these phenomena generalize to other models through preliminary studies with `Gemma 7B`.

Baseline Prompting In the baseline setup, the model is prompted with the query as is.

Prompt prefix (PP) We test the effects of inducing the user persona by adding a prompt prefix to the attack (PP). For each persona, we generated approximately 100 statements of identity (e.g., for the curious persona: *'I am motivated by the pursuit of knowledge.'*, *'I value intellectual integrity.'*, etc.). For PP, we prepend these statements to the attack prompt.

Activation steering via contrastive activation addition (CAA) For a given layer l, we create a persona steering vector following the contrastive activation addition (CAA) method [Rimsky et al., 2023]. The vector is derived by averaging the difference vectors between contrastive pairs of inputs X, Y where X exemplifies the behavior toward which we want to steer the model, and Y its opposite. We create these pairs as follows: we automatically rephrase each statement of persona identity as a yes or no question (e.g. *'I look for evidence to support my beliefs'* $\rightarrow$ *'Do I look for evidence to support my beliefs? Choices: (A) Yes. (B) No. Answer:'*), where the model is conditioned to either respond with *'(A)'* or *'(B)'* (we randomized the assignment of *'(A)'* and *'(B)'* to control for bias with respect to the label form). At inference time, we prompt the model with an adversarial query and calculate hidden representations until layer l, then add the steering vector to all positions in layer l and continue the model's forward computation from there. We study both adding the steering vector (CAA+) and subtracting it (CAA-) from the hidden representations. We consider every other layer between 5 and 39 (for efficiency) in `Llama 2 13B chat` (40 layers total) and select l based on the

[2]There are other types of restricted behaviors that may not necessarily generate harmful text, but are undesirable nonetheless, for example, hallucinating personal details. In this paper, we focus solely on a subset of harmful generations.

[3]`text-unicorn@001`, also known as the largest model in the PaLM family, is a foundation model optimized for a variety of language tasks and is available via the public Cloud Vertex AI API `https://cloud.google.com/vertex-ai/generative-ai/docs/model-reference/text`.

[4]The *AdvBench* attacks tend to be blatant, and without any adjustments to the prompts, we observed that the models under study would refuse them all.